Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Probability

SOLUTIONS

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1. $\frac{1}{2}$. **2.** $\frac{4}{52} = \frac{1}{13}$. **3.** 0.9. **4.** 0.18. **5.** 0.7.
6. $P(B|A) = \frac{0.2}{0.6} = \frac{1}{3}$. **7.** $\frac{1}{4}$. **8.** $\frac{5}{8}$.
9. $\frac{5}{8} \times \frac{4}{7} = \frac{5}{14}$. **10.** $\frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$. **11.** 0.3.
12. $0.4 + 0.5 - 0.2 = 0.7$. **13.** 0.4. **14.** 0.3. **15.** 0.75.
16. $P(\text{same}) = \frac{\binom{4}{2}+\binom{3}{2}+\binom{5}{2}}{\binom{12}{2}} = \frac{6+3+10}{66} = \frac{19}{66}$.
17. $P(A \cap B) = 0.3$; $P(A)P(B) = 0.3$ so independent.
18. $\frac{6\times5\times4}{6^3} = \frac{5}{9}$.
19. Paths AA, ABAA, BAAA etc.; sum geometric series ≈ 0.725.
20. $P(R) = \frac{1}{2}(\frac{3}{5} + \frac{4}{5}) = \frac{7}{10}$. **21.** $P(\text{bag1}|R) = \frac{3/10}{7/10} = \frac{3}{7}$.
22. $P(B) = 0.4 \times 0.3 + 0.6 \times 0.5 = 0.42$.
23. $P(A \cap B') = P(A)P(B') = P(A)(1 - P(B))$.
24. Sample space BB,BG,GB (exclude GG); $P(\text{both boys}) = \frac{1}{3}$.
25. $2k + k + \frac{2k}{3} = 1 \Rightarrow k = \frac{3}{11}$; $P(A) = \frac{6}{11}$.